

# Report

## TASK 1

### Introduction

The aim of this task is to implement KNN binary classification model completely from scratch and recommend the best value for K (number of neighbours) and distance matrix that produces the highest accuracy for test data.

We are given a CSV file “data.csv” containing the measurements from a digitized image of a fine needle aspirate (FNA) of a breast mass. We have to train the model to classify them into two diagnostic categories: Malignant (M) or Benign (B).

### Methodology & Experimental Setup

#### 1. Data Preprocessing

- The target variable "Diagnosis" was encoded into binary format (Malignant = 1, Benign = 0).
- The 'id' & 'Unnamed: 32' columns were removed because they are not a valid predictive features. Remaining attributes were isolated and utilized as the independent input features for the model.

- Standardization was applied to the 30 independent variables to ensure that features with larger magnitudes (e.g., Area) did not disproportionately influence the distance calculations.
- The data was divided into 80% training data and 20% test data.

## 2. KNN Implementation

The KNN classifier was implemented from scratch. For every test sample, the distance between the test sample and every training sample was calculated. The training samples were then sorted according to distance, and the K nearest samples were selected.

The predicted class was determined using majority voting among the selected neighbours.

The following distance metrics were implemented:

1. Euclidean Distance
2. Manhattan Distance
3. Minkowski Distance
4. Cosine Distance
5. Hamming Distance

The values of K tested were 3, 4, 9, 20, and 47.

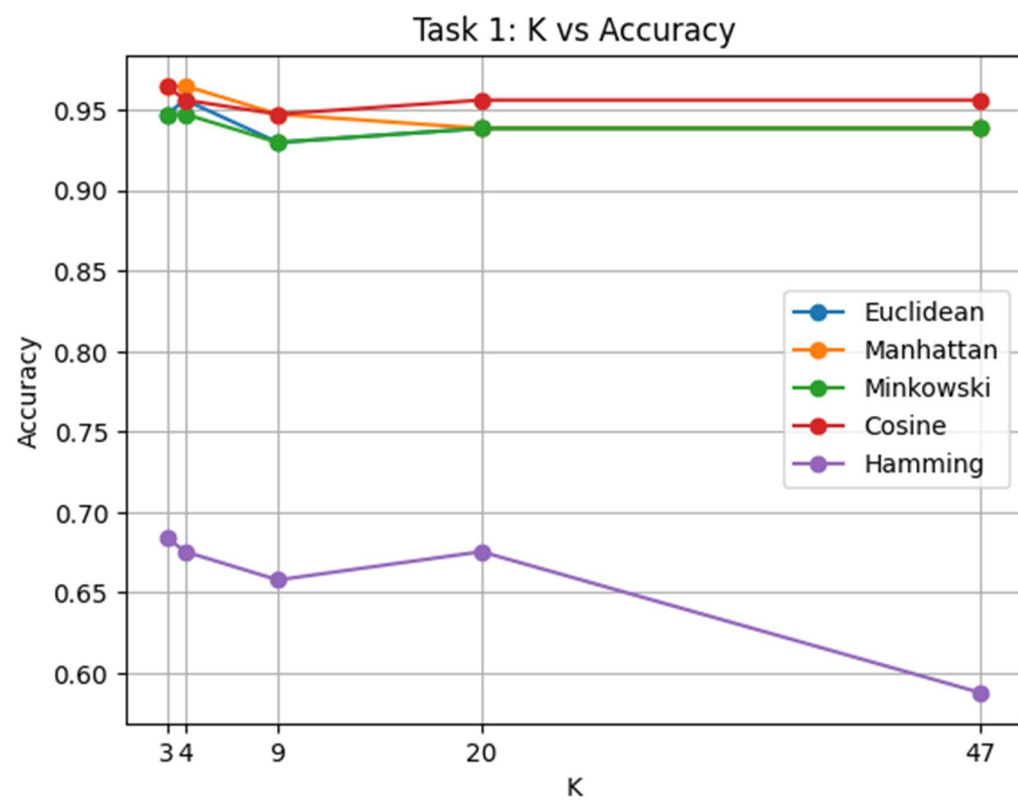
## Results

The testing accuracies obtained for Task 1 were:

| Distance Metric  | K = 3         | K = 4         | K = 9  | K = 20 | K = 47 |
|------------------|---------------|---------------|--------|--------|--------|
| Euclidean        | 94.74%        | 95.61%        | 92.98% | 93.86% | 93.86% |
| <b>Manhattan</b> | <b>96.49%</b> | <b>96.49%</b> | 94.74% | 93.86% | 93.86% |
| Minkowski        | 94.74%        | 94.74%        | 92.98% | 93.86% | 93.86% |
| <b>Cosine</b>    | 95.74%        | 95.61%        | 94.74% | 95.61% | 95.61% |
| Hamming          | 68.42%        | 67.54%        | 65.79% | 67.54% | 58.77% |

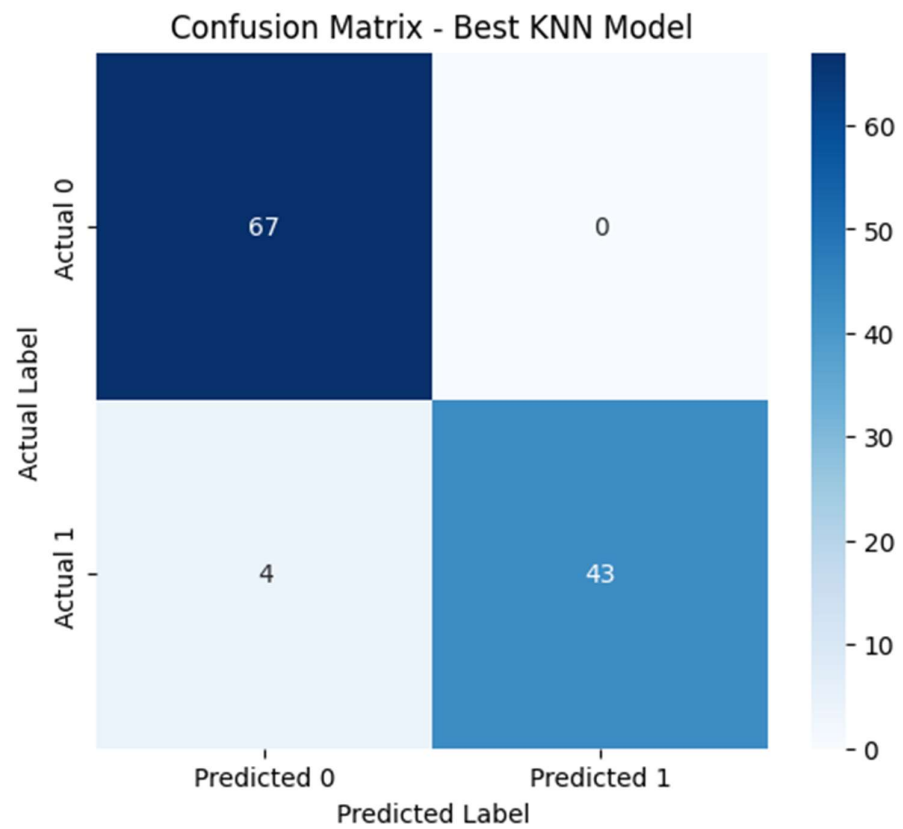
The best performing model for this task was :-

- Distance metric : Manhattan
- K = 3
- Testing Accuracy : 96.49%



## Confusion Matrix :

The confusion matrix for Manhattan distance with K = 3 was :



Therefore:

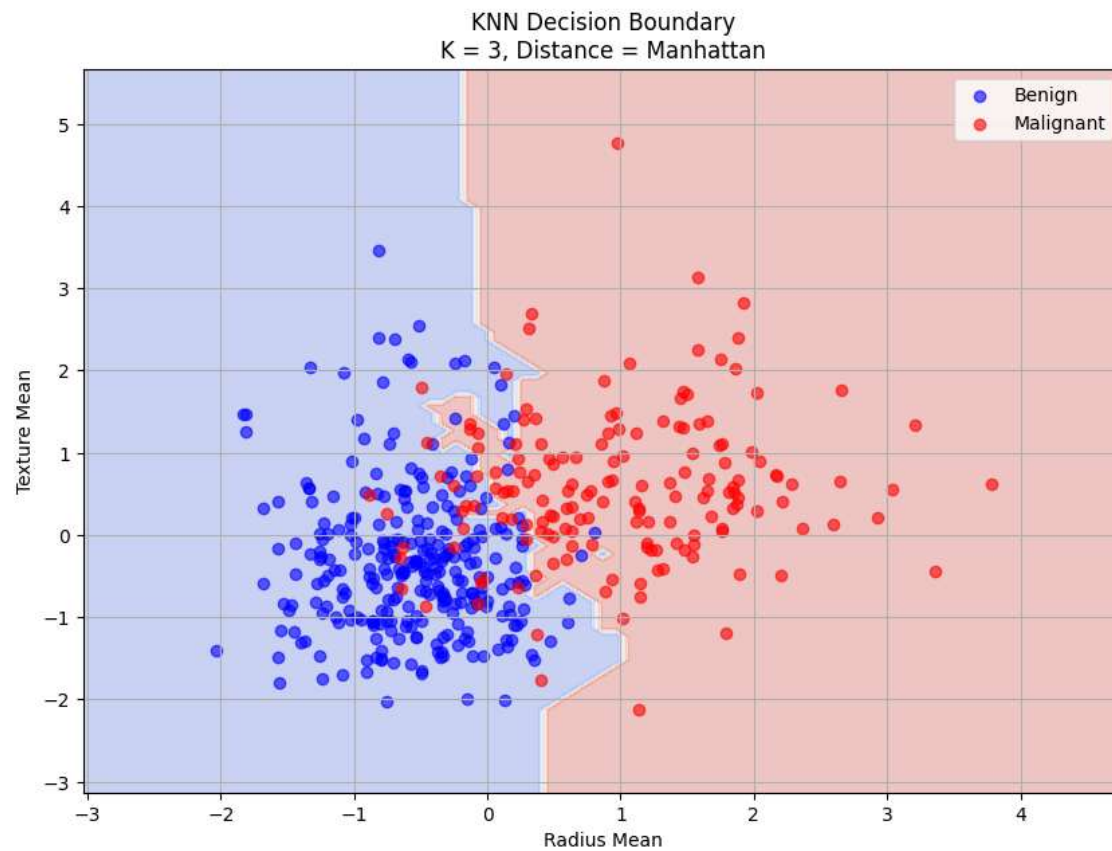
- True Negatives (TN) = 67
- False Positives (FP) = 0
- False Negatives (FN) = 4
- True Positives (TP) = 43

### Observations

1. Manhattan data performed the best. Accuracy of 96.49% was achieved using Manhattan distance with  $K = 3$ .
2. For most distance metrics, the accuracy was highest at  $K = 3$  or  $K = 4$ . Increasing  $K$  generally resulted in a decrease in performance.
3. Hamming distance produced accuracies between 58.77% and 68.42%, which were considerably lower than the other distance measures, possibly due to the fact that it is better suited for categorical data.
4. The choice of distance metric has a significant effect on KNN performance. Although the same training and testing data were used, the accuracy varied considerably depending on how similarity between samples was measured.
5. The selected model produced very few classification errors. Only four of the 114 test samples were misclassified.

### Bonus (Decision Boundary Visualization)

Visualization of the decision boundary for the best KNN model according to our results.





# TASK 2

## Introduction

The objective of this task is to apply a K-Nearest Neighbors (KNN) classification model built entirely from scratch to perform multi-class classification on the CIFAR-10 dataset. The model evaluates image inputs across 10 distinct classes: Airplane, Automobile, Bird, Cat, Deer, Dog, Frog, Horse, Ship, and Truck. The goal is to identify the optimal hyperparameter combination of neighbor count(K) and distance metric that achieves the highest test accuracy.

## Methodology & Experimental Setup

### 1. Data Preprocessing

- The CIFAR-10 dataset contains 60,000  $32 \times 32$  color images spanning 10 classes. Each image consists of  $32 \times 32 \times 3 = 3072$  pixel values.
- Computing pairwise distances between all test and training images in a scratch KNN implementation is computationally intensive. To make the evaluation tractable while preserving dataset variance, an experimental subset was constructed using 5,000 training images and 1,000 testing images.
- The pixel values were normalized from the range 0–255 to the range 0–1 by dividing by 255.

## 2. KNN Implementation

- The algorithm calculates vector distances between each test image vector and all training image vectors.
- Five distance metrics were tested: Euclidean, Manhattan, Minkowski ( $p=3$ ), Cosine, and Hamming.
- The instances are sorted by distance, the K-nearest samples are selected, and class assignment is determined via majority voting.
- The tested K values were 3, 4, 9, 20, and 47.

## Results

The testing accuracies obtained for Task 1 were :

| Distance Metric | K = 3  | K = 4  | K = 9             | K = 20 | K = 47 |
|-----------------|--------|--------|-------------------|--------|--------|
| Euclidean       | 28.20% | 28.40% | 28.40%            | 27.20% | 25.80% |
| Manhattan       | 30.30% | 30.90% | <b>**32.40%**</b> | 30.70% | 29.60% |
| Minkowski       | 23.70% | 23.70% | 25.90%            | 25.00% | 22.60% |
| Cosine          | 30.20% | 30.40% | 30.50%            | 29.30% | 27.60% |
| Hamming         | 23.20% | 25.20% | 25.10%            | 25.50% | 26.70% |

The best performing model for this task was :-

- **Distance Metric:** Manhattan
- **K:** 9
- **Testing Accuracy:** 32.40%

## Confusion Matrix

The confusion matrix for Manhattan distance with  $K = 9$  was:

| Actual \ Predicted | Airplane | Automobile | Bird | Cat | Deer | Dog | Frog | Horse | Ship | Truck |
|--------------------|----------|------------|------|-----|------|-----|------|-------|------|-------|
| Airplane           | 47       | 0          | 12   | 1   | 9    | 1   | 6    | 1     | 24   | 2     |
| Automobile         | 9        | 10         | 9    | 4   | 16   | 1   | 12   | 0     | 24   | 4     |
| Bird               | 15       | 0          | 49   | 1   | 19   | 4   | 6    | 1     | 5    | 0     |
| Cat                | 7        | 1          | 19   | 17  | 25   | 9   | 19   | 1     | 5    | 0     |
| Deer               | 6        | 1          | 18   | 5   | 41   | 1   | 8    | 1     | 8    | 1     |
| Dog                | 6        | 0          | 18   | 8   | 21   | 12  | 11   | 6     | 4    | 0     |

|              |    |   |    |   |    |   |    |    |    |    |
|--------------|----|---|----|---|----|---|----|----|----|----|
| <b>Frog</b>  | 2  | 0 | 27 | 6 | 42 | 3 | 27 | 2  | 2  | 1  |
| <b>Horse</b> | 5  | 0 | 20 | 4 | 27 | 4 | 11 | 20 | 9  | 2  |
| <b>Ship</b>  | 14 | 0 | 2  | 1 | 8  | 2 | 2  | 1  | 74 | 2  |
| <b>Truck</b> | 10 | 0 | 11 | 3 | 12 | 2 | 11 | 5  | 28 | 27 |

### Macro-metrics:

- **Macro Precision:** 41.83%
- **Macro Recall:** 32.02%

## Observations & Insights:

- Manhattan distance yielded the highest peak accuracy (32.40% at K=9), followed closely by Cosine distance (30.50% at K=9).
- The confusion matrix shows distinct clusters of error. Visually similar animal classes (cat, dog, deer, frog) frequently cross-predicted into each other due to pixel-level feature overlap. Conversely, structured vehicle backgrounds like water for ships enabled higher true positive performance (74 correct samples).
- The macro recall of 32.02% indicates that the model correctly identified approximately 32.02% of the actual samples across the classes on average.

## PLOT CIFR 10 K VS ACCURACY:

